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Department of Informatics, System and Communication (DISCo)

Master's Degree in Data Science

Predictive Analytics for Pension Preparedness: Machine Learning Customer Segmentation in the Italian Insurance Market

Data Science Lab Project



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July 21, 2025

Contents

1	Introduction	2
2	Dataset Description	2
3	Methodologies	3
3.1	Data Preprocessing and Advanced Feature Engineering	3
3.1.1	Target Variable Optimization	3
3.1.2	Comprehensive Feature Engineering Framework	4
3.1.3	Advanced Feature Selection Pipeline	4
3.1.4	Optimized Class Balancing Strategy	5
3.2	Advanced Machine Learning Algorithms	5
3.2.1	XGBoost Balanced	5
3.2.2	Random Forest	5
3.2.3	Logistic Regression	6
3.2.4	Gradient Boosting	6
3.2.5	Ensemble Voting	6
3.3	Dynamic Threshold Optimization	6
3.4	Comprehensive Model Evaluation	7
3.4.1	Cross-Validation Framework	7
3.4.2	Advanced Performance Metrics	7
4	Experiments	7
4.1	Experimental Design	8
4.2	Dataset Preparation	8
4.3	Preprocessing Pipeline	8
4.4	Algorithm Implementation and Parameter Selection	9
4.5	Threshold Optimization Strategy	9
4.6	Cross-Validation Strategy	9
5	Results	10
5.1	Comprehensive Performance Analysis	10
5.2	Individual Model Analysis	10
5.2.1	Logistic Regression	10
5.2.2	Random Forest	11
5.2.3	Ensemble Voting	11
5.2.4	XGBoost Balanced	12
5.2.5	Gradient Boosting	12
5.3	Strategic Trade-off Analysis	13
5.4	Business Application Scenarios	13
5.4.1	Target Coverage-Oriented Scenario	13
5.4.2	Balanced Trade-Off Scenario	13
5.4.3	Accuracy-Oriented Scenario	13
5.5	Implementation Considerations	14
6	Conclusions	14

1 Introduction

In Italy, the sustainability of the pension system is an increasingly pressing issue. The progressive ageing of the population, combined with a structural demographic decline, is putting the pay-as-you-go public pension model under pressure. The number of retirees is growing faster than the active workforce, making it inevitable, in the medium to long term, that the average pension benefit will decrease compared to the final salary earned during employment. This dynamic is further exacerbated by an uncertain economic landscape, marked by job instability, stagnant wages, and rising precariousness, particularly affecting young people and self-employed workers.

In this context, private pension schemes emerge as a key tool to ensure adequate income after retirement. However, in Italy, participation in such instruments, such as pension funds, individual pension plans (PIPs), or life insurance policies, remains limited and uneven, often influenced by socio-economic and cultural factors. Therefore, it becomes crucial for insurance companies to identify the population segments that are less prepared for retirement, and based on these insights, develop targeted communication and personalized offerings.

The objective of this project is to leverage data science techniques to support an Italian insurance company in improving its marketing strategies and client segmentation for pension-related products. Specifically, the aim is to identify groups of individuals at high risk of inadequate pension preparedness, by analyzing variables such as age, income, education level, geographic area, and financial behavior.

This analysis is intended to provide both strategic and operational value, helping the company to raise awareness among underprepared individuals and to expand access to complementary pension tools in a more inclusive and data-driven manner.

2 Dataset Description

The dataset used in this study originates from a comprehensive market survey conducted among clients of a national banking institution, made available to insurance companies for enhancing customer segmentation strategies in pension-related products. The survey, based on the 2023 IACOFI (International Adult Competency Framework on Financial Inclusion) questionnaire, encompasses demographic, socio-economic, and behavioral variables providing detailed insights into individuals' financial situations and attitudes.

The complete dataset includes 4,862 observations with 219 variables across multiple thematic domains. However, following data cleaning procedures that removed incomplete responses and invalid entries, the final analytical dataset comprised 4,553 observations. The target variable distribution reveals a significant class imbalance characteristic of financial literacy data: 3,505 samples (77%) classified as "High Financial Literacy" and 1,048 samples (23%) as "Low Financial Literacy".

The variables utilized in this study were strategically selected from seven key thematic areas:

- **Personal and demographic characteristics:** Gender, age, geographic area (North-West, North-East, Center, South, Islands), municipality size, family composition, education level, and citizenship status.
- **Employment and income:** Occupational status (employee, self-employed, unemployed, retired), work sector classification, and self-reported income brackets.
- **Financial behavior and responsibility:** Money management patterns, financial decision-making authority, and budgeting practices.
- **Financial resilience indicators:** Ability to cope with unexpected expenses, strategies employed when income is insufficient, and self-sufficiency duration without external borrowing.
- **Economic stress factors:** Difficulty covering expenses, financial stress levels, and economic vulnerability indicators.
- **Financial product awareness:** Knowledge and ownership of pension plans, savings accounts, investment products, and insurance policies.
- **Motivational and attitudinal factors:** Purchase motivations, risk preferences, long-term planning orientation, and confidence in financial decision-making.

The final feature set consists of 29 carefully selected variables that balance predictive power with practical data collection feasibility in insurance business contexts. These variables encompass both original survey responses and sophisticated engineered features designed to capture complex financial behavior patterns and demographic interactions relevant to pension preparedness assessment.

3 Methodologies

This section presents the comprehensive machine learning framework developed for financial literacy prediction, incorporating advanced preprocessing techniques, sophisticated feature engineering, and optimized algorithmic approaches specifically designed for minority class identification in imbalanced datasets.

3.1 Data Preprocessing and Advanced Feature Engineering

3.1.1 Target Variable Optimization

The target variable transformation process involved converting the original 5-point financial literacy scale into a strategically defined binary classification

framework. Responses of 1-2 on the original scale were classified as “Low Financial Literacy” (class 0), while responses of 3-5 were classified as “High Financial Literacy” (class 1). This transformation was specifically designed to identify customers requiring educational interventions and pension preparedness support, creating a practical business segmentation approach with clear actionable implications.

3.1.2 Comprehensive Feature Engineering Framework

The feature engineering process implemented a multi-layered approach generating 29 sophisticated variables from the original dataset. The engineering strategy followed four primary directions:

Demographic Segmentation: Age categorization into five strategic groups (Very Young: <25, Young Adult: 25-35, Middle Age: 35-55, Senior: 55+) designed to capture life-stage financial behavior patterns. Additional binary indicators were created for each age category to enable granular analysis of age-related financial literacy patterns.

Financial Behavior Indicators: Binary variables capturing critical financial responsibility patterns, including sole financial responsibility (Is_Sole_Responsible), shared financial management (Is_Shared_Responsible), and economic stress indicators (High_Financial_Difficulty, No_Financial_Difficulty).

Socioeconomic Classification: Income and education level binary indicators (Is_High_Income, Is_Low_Income, Is_High_Education, Is_Low_Education) using optimized thresholds determined through iterative analysis to maximize discriminative power for financial literacy prediction.

Advanced Composite Features: Sophisticated mathematical combinations capturing complex relationships:

$$\text{Education_Income_Ratio} = \frac{\text{Education Level}}{\text{Income Bracket} + 1} \quad (1)$$

$$\text{Age_Education_Product} = \text{Age} \times \text{Education Level} \quad (2)$$

$$\text{Financial_Stress_Score} = \text{High Financial Difficulty} \times 2 + \text{Is Low Income} \quad (3)$$

Interaction Terms: Strategic feature interactions including Young_High_Ed, Senior_High_Income, and Sole_Resp_High_Ed to capture demographic-behavioral intersections critical for financial literacy assessment.

3.1.3 Advanced Feature Selection Pipeline

A sophisticated two-stage feature selection process was implemented to optimize dimensionality while maximizing minority class detection capability:

Stage 1 - Statistical Selection: SelectKBest with f_{classif} scoring reduced the initial 32 encoded features to the 25 most statistically significant variables using the F-statistic:

$$F = \frac{\text{MSB}}{\text{MSW}} \quad (4)$$

where MSB represents between-group variance and MSW represents within-group variance.

Stage 2 - Recursive Feature Elimination: RFE using Random Forest as the base estimator further refined the selection to 20 optimal features through iterative elimination of least important variables. The final selected feature indices were: [0, 1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 15, 16, 17, 20, 21, 22, 23].

3.1.4 Optimized Class Balancing Strategy

The class imbalance challenge was addressed through an aggressive two-step resampling approach specifically optimized for minority class detection:

SMOTE Augmentation: Synthetic Minority Oversampling Technique with optimized parameters: - $k_neighbors = 3$ for maximum synthetic sample diversity - Aggressive sampling strategy increasing class 0 representation to 90% of class 1 frequency - Synthetic sample generation using: $x_{new} = x_i + \lambda \times (x_{neighbor} - x_i)$ where $\lambda \in [0, 1]$

Conservative UnderSampling: Random UnderSampling with minimal majority class reduction (ratio 1.05:1) to preserve essential class 1 patterns while achieving near-balanced training data.

The final resampling results achieved: - Original distribution: Class 0: 838 samples, Class 1: 2,804 samples - Post-SMOTE: Class 0: 2,523 samples, Class 1: 2,804 samples - Final balanced: Class 0: 2,523 samples (48.8%), Class 1: 2,649 samples (51.2%)

3.2 Advanced Machine Learning Algorithms

3.2.1 XGBoost Balanced

XGBoost was configured with parameters specifically optimized for minority class detection while maintaining computational efficiency:

Core Parameters: $n_estimators = 180$, $max_depth = 4$, $learning_rate = 0.07$, with minimal regularization ($\alpha = \lambda = 0.03$) to enhance sensitivity to minority class patterns.

Class Balancing: Moderate $scale_pos_weight$ (1.5:1) balancing minority class enhancement with overall model stability.

The objective function optimization follows:

$$\mathcal{L}(\phi) = \sum l(y_i, \hat{y}_i) + \sum \Omega(f_k) \quad (5)$$

with regularization term: $\Omega(f) = \gamma T + \frac{1}{2} \lambda ||w||^2$

3.2.2 Random Forest

Random Forest implementation utilized ensemble diversity optimization:

Ensemble Configuration: $n_estimators = 250$, $max_depth = 9$, $min_samples_split = 7$, $min_samples_leaf = 3$, with $max_features = 'sqrt'$ for optimal feature randomness.

Class Weighting: Custom weights (1.5:1 ratio) favoring minority class identification while maintaining ensemble robustness.

The ensemble prediction follows: $\hat{y} = \frac{1}{B} \sum_{b=1}^B T_b(x)$ with Gini impurity splitting criterion.

3.2.3 Logistic Regression

Logistic Regression was implemented with aggressive minority class optimization:

Regularization: L2 penalty with $C = 1.0$ providing moderate regularization, $max_iter = 3000$ ensuring convergence.

Aggressive Class Weighting: 3.0:1 ratio maximizing minority class sensitivity, representing the most aggressive class balancing approach among all models.

The log-odds formulation: $\log\left(\frac{p}{1-p}\right) = \beta_0 + \sum_{i=1}^k \beta_i x_i$

3.2.4 Gradient Boosting

Standard Gradient Boosting with balanced parameters:

Sequential Learning: $n_estimators = 150$, $learning_rate = 0.08$, $max_depth = 4$, $subsample = 0.85$ for regularization.

The sequential ensemble construction: $F(x) = \sum_{m=1}^M \gamma_m h_m(x)$ with gradient-based residual fitting.

3.2.5 Ensemble Voting

Advanced ensemble approach combining probabilistic outputs from top-performing individual models using optimized threshold selection specifically designed for minority class maximization.

3.3 Dynamic Threshold Optimization

A critical innovation in the methodology was the implementation of dynamic threshold optimization specifically designed to maximize true negative identification (customers with low financial literacy). The optimization process evaluated thresholds ranging from 0.05 to 0.7 using a sophisticated composite scoring function:

$$\text{Score} = \begin{cases} 0.8 \times \text{Sensitivity}_{class_0} + 0.15 \times \text{Accuracy} + 0.05 \times \text{Specificity} & \text{if TN} \geq 150 \\ 0.7 \times \text{Sensitivity}_{class_0} + 0.2 \times \text{Accuracy} + 0.1 \times \text{Specificity} & \text{if TN} \geq 120 \\ 0.4 \times \text{Sensitivity}_{class_0} + 0.3 \times \text{Accuracy} + 0.3 \times \text{Specificity} & \text{otherwise} \end{cases} \quad (6)$$

The thresholds of 150 and 120 True Negatives are defined based on their relative numerosity within the test dataset (total: 911 samples). This stratified approach enables quantification of the trade-off between metrics as a function of the model's actual capacity to identify the target segment:

- **TN $\geq$ 150 (16.5% of test set):** Represents excellent performance in target identification, justifying maximum priority to sensitivity (weight 0.8)
- **TN $\geq$ 120 (13.2% of test set):** Indicates good but sub-optimal performance, requiring greater balance with accuracy (weights 0.7 vs 0.2)
- **TN $<$ 120:** Performance insufficient for target-only prioritization, necessitating equilibrium across all metrics (balanced weights)

This segmentation rewards models achieving operationally significant target identification thresholds, providing management with clear indicators for cost-benefit evaluation of different implementation strategies while ensuring maximum identification of the target segment through performance-based metric weighting.

3.4 Comprehensive Model Evaluation

3.4.1 Cross-Validation Framework

Model evaluation employed 10-fold stratified cross-validation with fixed random state (42) ensuring reproducibility. The stratification maintained identical class distributions across all folds, preventing evaluation bias in imbalanced datasets.

3.4.2 Advanced Performance Metrics

Beyond standard classification metrics, specialized measures were implemented for minority class evaluation:

Standard Metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (10)$$

4 Experiments

This section details the experimental setup, parameter choices, and implementation details necessary for reproducibility of the results, with a specific focus on optimizing the identification of customers with low financial literacy.

4.1 Experimental Design

The experimental framework was designed to compare five different machine learning algorithms for predicting financial literacy levels in a binary classification task. The primary objective was to evaluate multiple algorithmic approaches for detecting customers with low financial literacy who represent the key target segment for pension education and preparedness programs, analyzing the trade-offs between different performance metrics and operational considerations.

4.2 Dataset Preparation

The original dataset contained 4,553 samples with 29 selected features after preprocessing. The target variable distribution showed a significant class imbalance with 3,505 samples (77%) labeled as “High Financial Literacy” and 1,048 samples (23%) as “Low Financial Literacy”. The dataset was split using stratified sampling with 80% for training (3,642 samples) and 20% for testing (911 samples), ensuring proportional representation of both classes in each subset.

4.3 Preprocessing Pipeline

The preprocessing pipeline consisted of four main stages designed to address class imbalance while preserving data integrity. First, categorical variables were encoded using one-hot encoding with the `drop_first=True` parameter to avoid multicollinearity, resulting in 32 encoded features. Second, a two-stage feature selection process was implemented: `SelectKBest` with `f_classif` scoring reduced the feature space to the 25 most informative variables, followed by `Recursive Feature Elimination (RFE)` that further refined the selection to 20 optimal features. Third, standardization was applied using `StandardScaler` for algorithms sensitive to feature scales. Fourth, class balancing was implemented through a strategic resampling approach.

The comprehensive feature engineering process generated 29 variables from the original dataset, encompassing fundamental customer attributes (Gender, Geographic_Area, Age, Education_Level, Work_Sector, Citizenship, Income_Bracket, Purchase_Motivation, Money_responsible) and advanced composite indicators (Age_Group, Is_Very_Young, Is_Young_Adult, Is_Middle_Age, Is_Senior, Is_Sole_Responsible, Is_Shared_Responsible, High_Financial_Difficulty, No_Financial_Difficulty, Is_High_Income, Is_Low_Income, Is_High_Education, Is_Low_Education, Education_Income_Ratio, Age_Education_Product, Financial_Stress_Score, Young_High_Ed, Senior_High_Income, Sole_Resp_High_Ed, difficulty_covering_expenses).

The resampling strategy employed `SMOTE` to generate synthetic samples for the minority class, with optimized parameters: k neighbors equal to 3 for diversity and explicit sampling to increase class 0 representation to 90% of class 1 frequency. This approach increased the minority class from 838 to 2,523 samples (201% increase), while the majority class was reduced from 2,804 to 2,649

samples through Random UnderSampling, creating a nearly balanced training set (48.8% vs 51.2%).

4.4 Algorithm Implementation and Parameter Selection

Each algorithm was implemented with carefully selected hyperparameters, with different approaches to class weighting and regularization to explore various solutions to the class imbalance challenge.

XGBoost was configured with $n_estimators = 180$, $max_depth = 4$, $learning_rate = 0.07$, regularization parameters $\alpha = \lambda = 0.03$, and moderate class weighting (ratio 1.5:1) to balance performance across classes.

Random Forest employed $n_estimators = 250$ trees, $max_depth = 9$, $min_samples_split = 7$, $min_samples_leaf = 3$, $max_features = 'sqrt'$, and moderate class weighting (ratio 1.5:1) for robust ensemble learning.

Gradient Boosting utilized $n_estimators = 150$, $learning_rate = 0.08$, $max_depth = 4$, $subsample = 0.85$, and standard parameters optimized for gradient-based learning.

Logistic Regression was implemented with L2 regularization, $C = 1.0$, and aggressive class weighting (ratio 3.0:1) to maximize sensitivity to the minority class.

The Ensemble Voting approach combined probabilistic outputs from multiple models using threshold optimization.

4.5 Threshold Optimization Strategy

Dynamic threshold optimization was implemented to evaluate different performance trade-offs. For models with probabilistic output, thresholds ranging from 0.05 to 0.7 were systematically evaluated using a composite scoring function:

$$\text{Score} = \begin{cases} 0.8 \times \text{Sensitivity}_{class_0} + 0.15 \times \text{Accuracy} + 0.05 \times \text{Specificity} & \text{if TN} \geq 150 \\ 0.7 \times \text{Sensitivity}_{class_0} + 0.2 \times \text{Accuracy} + 0.1 \times \text{Specificity} & \text{if TN} \geq 120 \\ 0.4 \times \text{Sensitivity}_{class_0} + 0.3 \times \text{Accuracy} + 0.3 \times \text{Specificity} & \text{otherwise} \end{cases}$$

This approach enabled evaluation of different strategic priorities in model selection.

4.6 Cross-Validation Strategy

Model evaluation employed 10-fold stratified cross-validation with fixed random state (42) for reproducibility. Multiple metrics were computed including accuracy, F1-score, and class-specific performance measures to provide comprehensive evaluation across different business scenarios.

5 Results

This section presents a comprehensive analysis of the experimental results, examining the performance characteristics and trade-offs of each algorithmic approach for customer segmentation in financial literacy identification.

5.1 Comprehensive Performance Analysis

The experimental evaluation revealed distinct performance profiles across the five machine learning algorithms, each demonstrating different strengths and limitations for financial customer segmentation applications.

Table 1 presents the complete performance analysis across all evaluated models:

Table 1: Comprehensive Performance Analysis Across All Models

Metric	Logistic Reg.	Random Forest	Ensemble	XGBoost	G. Boost.
Test Accuracy	24.4%	42.3%	40.6%	55.5%	60.9%
CV Accuracy	55.3%	80.5%	N/A	81.9%	81.6%
Low FL Recall	97.6%	77.1%	75.2%	48.1%	43.3%
Low FL Precision	23.1%	25.3%	24.4%	25.4%	27.7%
True Negatives	205	162	158	101	91
False Positives	5	48	52	109	119
Custom Score	0.866	0.719	0.700	0.500	0.477

5.2 Individual Model Analysis

5.2.1 Logistic Regression

Strengths:

- Exceptional minority class identification with 97.6% recall for low financial literacy customers
- Minimal target loss with only 5 customers missed out of 210 (2.4% missed opportunity rate)
- High interpretability and operational simplicity for production deployment
- Optimal threshold of 0.69 provides clear decision boundary
- Highest custom score (0.866) when prioritizing target identification

Limitations:

- Lowest overall test accuracy at 24.4% due to aggressive minority class focus
- High false negative rate (684 customers) leading to potential over-targeting in educational campaigns

- Poor performance on majority class identification (2.4% high FL recall)
- Significant resource allocation implications due to broad targeting approach

Optimal Use Cases: Maximum coverage educational campaigns where missing target customers is more costly than over-inclusion.

5.2.2 Random Forest

Strengths:

- Strong minority class performance with 77.1% low FL recall
- Balanced approach with reasonable overall accuracy (42.3%)
- Robust ensemble methodology reducing overfitting risk
- Good cross-validation stability (80.5% CV accuracy)
- Moderate false positive rate (48 customers) for efficient targeting

Limitations:

- Misses 48 target customers (22.9% of low FL segment)
- Lower precision compared to tree-based alternatives
- Complex model interpretation compared to linear approaches
- Computational overhead for real-time applications

Optimal Use Cases: Balanced educational campaigns requiring reasonable accuracy across both customer segments.

5.2.3 Ensemble Voting

Strengths:

- Combines strengths of multiple algorithms for robust predictions
- Good minority class recall (75.2%) with acceptable overall accuracy (40.6%)
- Potential for improved generalization through ensemble diversity
- Flexible threshold optimization for different business priorities

Limitations:

- Increased complexity in deployment and maintenance
- Moderate target loss (52 customers missed)
- Computational overhead from multiple model execution
- Limited interpretability due to ensemble nature

Optimal Use Cases: Applications requiring robust performance across varying data conditions with acceptable complexity trade-offs.

5.2.4 XGBoost Balanced

Strengths:

- Strong overall accuracy performance (55.5%)
- Excellent cross-validation stability (81.9% CV accuracy)
- Advanced gradient boosting methodology for complex pattern detection
- Good precision for low FL identification (25.4%)
- Reasonable computational efficiency for production deployment

Limitations:

- Moderate minority class recall (48.1%) missing significant target population
- High false positive rate (109 customers) indicating substantial target loss
- Complex hyperparameter tuning requirements
- Less interpretable than linear models for regulatory compliance

Optimal Use Cases: Scenarios prioritizing overall accuracy and computational efficiency over maximum target coverage.

5.2.5 Gradient Boosting

Strengths:

- Highest overall test accuracy (60.9%) among all models
- Excellent cross-validation performance (81.6% CV accuracy)
- Strong performance on majority class identification
- Highest precision for low FL customers (27.7%)
- Stable gradient-based learning approach

Limitations:

- Lowest minority class recall (43.3%) missing majority of target segment
- Highest false positive rate (119 customers) for educational targeting
- Focus on overall performance compromises target identification
- Sequential training limitations for parallel processing

Optimal Use Cases: Applications where overall accuracy is prioritized over minority class identification, such as general customer analytics.

5.3 Strategic Trade-off Analysis

The experimental results reveal fundamental trade-offs between different performance objectives:

Table 2: Strategic Trade-off Analysis Across Models (Quantitative Summary)

Model	Target Coverage (Recall)	False Positives (FP)	Overall Accuracy
Logistic Regression	97.6%	684	24.4%
Random Forest	77.1%	48	42.3%
Ensemble Voting	75.2%	52	40.6%
XGBoost	48.1%	109	55.5%
Gradient Boosting	43.3%	119	60.9%

5.4 Business Application Scenarios

The following scenarios illustrate different observed model behaviors based on distinct performance profiles. Each scenario is described without interpretative judgment or recommendation.

5.4.1 Target Coverage-Oriented Scenario

The Logistic Regression model achieves a recall of 97.6% on the minority class (low financial literacy), corresponding to the identification of nearly all individuals in the target group. This comes with an overall test accuracy of 24.4% and a high number of false positives (684), resulting in broader classification of non-target individuals.

5.4.2 Balanced Trade-Off Scenario

The Random Forest (77.1% recall, 42.3% accuracy) and Ensemble Voting (75.2% recall, 40.6% accuracy) models show intermediate performance. They identify a substantial portion of the target class while reducing both false positives and false negatives compared to Logistic Regression. These results reflect a trade-off between target coverage and classification precision.

5.4.3 Accuracy-Oriented Scenario

XGBoost (55.5% accuracy, 48.1% recall) and Gradient Boosting (60.9% accuracy, 43.3% recall) obtain the highest overall classification accuracy. These models are associated with lower sensitivity on the minority class and higher counts of false negatives (109 and 119 missed target individuals, respectively), focusing performance on overall correctness rather than specific subgroup detection.

5.5 Implementation Considerations

Each model evaluated in this study exhibits distinct characteristics with respect to implementation requirements:

Interpretability: Logistic Regression enables transparent interpretation of coefficients and decision boundaries, which may facilitate communication in regulated environments.

Computational Requirements: Gradient Boosting and XGBoost demonstrate faster training and inference times in high-volume settings, due to their optimized implementations and support for parallel processing.

Maintenance and Tuning: Linear models such as Logistic Regression generally involve fewer hyperparameters and require less tuning effort compared to ensemble methods, which involve more complex parameter grids and interactions.

Scalability: Tree-based models such as Random Forest, XGBoost, and Gradient Boosting show stable performance with increasing dataset size and feature dimensionality, making them applicable to larger-scale segmentation tasks.

6 Conclusions

The experimental analysis highlights how different algorithmic approaches to customer segmentation for financial literacy identification are associated with varying performance profiles, which can be interpreted in light of diverse business priorities and operational constraints.

The results suggest three possible strategic orientations, each corresponding to specific model characteristics:

Target Coverage-Oriented Approach: The Logistic Regression model achieves high recall on the minority class (97.6%), allowing for the identification of most individuals potentially in need of financial education. This outcome is associated with lower overall accuracy (24.4%) and broader inclusion of non-target individuals.

Balanced Trade-off Approach: The Random Forest and Ensemble Voting models demonstrate an intermediate profile, combining recall rates between 75–77% with moderate levels of overall accuracy and false positive rates. This configuration reflects a balance between inclusiveness and resource allocation.

Accuracy-Oriented Approach: XGBoost and Gradient Boosting obtain higher overall classification accuracy (55–61%) while presenting lower recall on the minority class (43–48%), suggesting a potential fit for applications in which global predictive performance is prioritized over specific subgroup coverage.

The feature selection procedure, which reduced the input space from 32 to 20 features, showed consistent predictive utility across all algorithms. All selected features belong to categories relevant and accessible within insurance business contexts, supporting practical applicability.

Dynamic threshold optimization allowed further exploration of model behavior, showing how trade-offs between sensitivity, accuracy, and specificity can be

managed to align with different strategic objectives. The class imbalance issue was addressed through a combined SMOTE and undersampling strategy, resulting in a balanced training set that improved minority class detection across all models.

The presented framework does not aim to define a unique optimal solution but rather to provide a comparative basis for potential deployment paths. The analysis enables decision-makers to assess model behaviors according to their own priorities in terms of coverage, efficiency, and accuracy.